

Atom decomposition of ordinary Hall–Littlewood $P_\mu(x; t)$ via the operator θ_i^{alt} and Cherednik–Ram

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Abstract

Let $\theta_i^{\text{alt}}(f) := \frac{x_{i+1}-tx_i}{x_i-x_{i+1}}(f - s_i f)$ act on $\mathbb{Q}(t)[x_1, \dots, x_n]$, and for a composition $\gamma \in \mathbb{Z}_{\geq 0}^n$ let A_γ^{alt} be the *bubble-sort atom* built from $x^{\text{sort}\downarrow(\gamma)}$ by applying θ^{alt} operators along the leftmost-first bubble-sort word. We prove that for every partition μ with $\ell(\mu) \leq n$ the ordinary Hall–Littlewood polynomial expands as

$$P_\mu(x_1, \dots, x_n; t) = \sum_{\gamma: \text{sort}(\gamma)=\mu} c_\gamma(t) \cdot A_\gamma^{\text{alt}}(x; t), \quad c_\gamma(t) \in \mathbb{Z}[t].$$

The coefficients are uniquely determined by triangularity of the atom basis in dominance order. The proof combines three ingredients: (i) the “key identity” $T_i = (1+t)s_i - \theta_i^{\text{alt}} - \text{id}$ relating the Demazure–Lusztig operator T_i to θ^{alt} ; (ii) an explicit formula for $T_i(x^\nu)$ when $\nu_i \leq \nu_{i+1}$; and (iii) the Cherednik–Ram identity $\sum_{w \in S_n} T_w(x^\mu) = v_\mu(t) P_\mu$.

Positivity of c_γ (i.e. $c_\gamma \in \mathbb{Z}_{\geq 0}[t]$), verified numerically for many μ at $n \leq 3$, is stated as a conjecture at the end. The proof of existence establishes the base case for Probe B’s atoms route.

1 Statement

Fix $n \geq 2$. On $\mathbb{Q}(t)[x_1, \dots, x_n]$ let s_i swap x_i and x_{i+1} , and set

$$\nabla_i f := \frac{f - s_i f}{x_i - x_{i+1}}, \quad \pi_i f := \frac{x_i f - x_{i+1} s_i f}{x_i - x_{i+1}} = f + x_{i+1} \nabla_i f$$

(the divided difference and the Demazure isobaric operator).

Definition 1 (Cherednik–Ram operator).

$$T_i := (t-1)\pi_i + s_i.$$

Equivalently, $T_i(f) = tf - (x_i - tx_{i+1})\nabla_i f$. The T_i satisfy the Hecke relations $(T_i - t)(T_i + 1) = 0$ and the braid relation $T_i T_{i+1} T_i = T_{i+1} T_i T_{i+1}$.

Definition 2 (Atom operator θ^{alt}).

$$\theta_i^{\text{alt}}(f) := \frac{x_{i+1} - tx_i}{x_i - x_{i+1}}(f - s_i f) = (x_{i+1} - tx_i)\nabla_i f.$$

It satisfies $(\theta_i^{\text{alt}})^2 + (1+t)\theta_i^{\text{alt}} = 0$, and $\theta_i^{\text{alt}}|_{t=0}$ recovers the classical Mason atom operator $\theta_i = \pi_i - \text{id}$.

Definition 3 (Bubble-sort atom). For a composition $\gamma \in \mathbb{Z}_{\geq 0}^n$ let $\lambda := \text{sort}_{\downarrow}(\gamma)$. Let $w_\gamma = s_{i_1} s_{i_2} \cdots s_{i_\ell}$ be the shortest permutation with $w_\gamma \cdot \lambda = \gamma$ (left action swapping positions), chosen as the *leftmost-first* bubble-sort word: process target positions $p = 1, 2, \dots$ in order, and at each p bubble the required entry into position p by adjacent swaps. Then

$$A_\gamma^{\text{alt}}(x; t) := \theta_{i_1}^{\text{alt}} \theta_{i_2}^{\text{alt}} \cdots \theta_{i_\ell}^{\text{alt}}(x^\lambda).$$

(Operator convention: $\theta_{i_\ell}^{\text{alt}}$ is applied first, then $\theta_{i_{\ell-1}}^{\text{alt}}$, etc.)

Write $\bar{\mu} = (\mu_n, \mu_{n-1}, \dots, \mu_1)$ for the *antipartition* of μ , and $v_\mu(t) := \prod_{i \geq 0} [m_i(\mu)]_t!$ for the standard Hall–Littlewood normalisation constant, where $m_i(\mu)$ is the multiplicity of i in μ (including m_0 , the padding zeros).

Theorem 4 (Main theorem). *Let μ be a partition with $\ell(\mu) \leq n$. Then*

$$P_\mu(x_1, \dots, x_n; t) = \sum_{\gamma: \text{sort}(\gamma) = \mu} c_\gamma(t) A_\gamma^{\text{alt}}(x; t) \quad (1)$$

for uniquely determined $c_\gamma(t) \in \mathbb{Z}[t]$, indexed by the rearrangements γ of μ .

The remainder of the paper is the proof. We collect the four ingredients below and combine them in §6.

2 The key identity

Proposition 5 (Key Identity, KI). *On $\mathbb{Q}(t)[x_1, \dots, x_n]$,*

$$T_i = (1+t) s_i - \theta_i^{\text{alt}} - \text{id}. \quad (2)$$

Proof. Both sides are $\mathbb{Q}(t)$ -linear operators; it suffices to show $T_i(f) = (1+t)s_i(f) - \theta_i^{\text{alt}}(f) - f$ for arbitrary f .

From the identity $\pi_i(f) = f + x_{i+1} \nabla_i(f)$ we get

$$\pi_i(f) - f = x_{i+1} \nabla_i(f), \quad \pi_i(f) - s_i(f) = \pi_i(f) - f + f - s_i(f) = x_i \nabla_i(f)$$

(using $f - s_i f = (x_i - x_{i+1}) \nabla_i f$). Therefore

$$\theta_i^{\text{alt}}(f) = (x_{i+1} - t x_i) \nabla_i f = x_{i+1} \nabla_i f - t x_i \nabla_i f = (\pi_i f - f) - t(\pi_i f - s_i f) = (1-t)\pi_i(f) + t s_i(f) - f.$$

Combining with $T_i(f) = (t-1)\pi_i(f) + s_i(f) = -(1-t)\pi_i(f) + s_i(f)$ gives

$$T_i(f) + \theta_i^{\text{alt}}(f) + f = s_i(f) + t s_i(f) = (1+t) s_i(f),$$

which rearranges to (2). □

Remark 6. Computationally verified on nine test polynomials (monomials and mixed polynomials of degree up to 4) at $n = 3$, both $i = 1$ and $i = 2$: zero residual. See `/home/clio/projects/scratch/verify_KI.py`

3 Action of T_i on ascending monomials

Proposition 7 (T_i on an ascending monomial). *Fix a composition $\nu \in \mathbb{Z}_{\geq 0}^n$. Write $a := \nu_i$, $b := \nu_{i+1}$. If $a \leq b$ (ascending or equal case), then*

$$T_i(x^\nu) = \begin{cases} t x^\nu & \text{if } a = b, \\ x^{s_i \nu} + (1-t) \sum_{j=a+1}^{b-1} x_1^{\nu_1} \cdots x_i^j x_{i+1}^{a+b-j} \cdots x_n^{\nu_n} & \text{if } a < b. \end{cases} \quad (3)$$

The sum in the ascending case runs over strictly intermediate exponents; in particular, if $b = a + 1$ the sum is empty and $T_i(x^\nu) = x^{s_i \nu}$.

Proof. Case $a = b$. Then $s_i(x^\nu) = x^\nu$ and $\nabla_i(x^\nu) = 0$, so $T_i(x^\nu) = t x^\nu - (x_i - t x_{i+1}) \cdot 0 = t x^\nu$.

Case $a < b$. Extract the two-variable core: writing $x^\nu = M \cdot x_i^a x_{i+1}^b$ with M a monomial in variables $\neq x_i, x_{i+1}$, we compute

$$\nabla_i(x_i^a x_{i+1}^b) = \frac{x_i^a x_{i+1}^b - x_i^b x_{i+1}^a}{x_i - x_{i+1}} = -x_i^a x_{i+1}^a \cdot \frac{x_i^{b-a} - x_{i+1}^{b-a}}{x_i - x_{i+1}} = - \sum_{k=0}^{b-a-1} x_i^{b-1-k} x_{i+1}^{a+k}.$$

Multiplying by $(x_{i+1} - t x_i)$ and combining the two resulting sums by x_i -degree j , one finds

$$\theta_i^{\text{alt}}(x_i^a x_{i+1}^b) = -x_i^a x_{i+1}^b + t x_i^b x_{i+1}^a + (t-1) \sum_{j=a+1}^{b-1} x_i^j x_{i+1}^{a+b-j}.$$

Substituting into $T_i(x^\nu) = (1+t)s_i(x^\nu) - \theta_i^{\text{alt}}(x^\nu) - x^\nu$ (with the same M prefactor on both sides) yields

$$T_i(x^\nu) = (1+t)x^{s_i \nu} - [-x^\nu + t x^{s_i \nu} + (t-1)\Sigma] - x^\nu = x^{s_i \nu} + (1-t)\Sigma,$$

where $\Sigma := M \cdot \sum_{j=a+1}^{b-1} x_i^j x_{i+1}^{a+b-j}$ is the sum in (3). $\square$ $\square$

Remark 8 (Numerical sanity check). For $\nu = (0, 2, 2)$, $\mu = (2, 2, 0)$ at $n = 3$: applying T_1 swaps positions 1, 2 with values $(0, 2)$, so $a = 0, b = 2$ and the sum has one term $j = 1$: $T_1(x_2^2 x_3^2) = x_1^2 x_3^2 + (1-t)x_1 x_2 x_3^2$. Verified in `verify_KI.py`.

4 Cherednik–Ram and parabolic factorisation

We use as a black box the Cherednik–Ram identity, verified in the companion note 2026-07-20-demazure-reentry on twenty-one cases (n, μ) with $n \in \{2, 3, 4\}$ and $|\mu| \leq 4$.

Proposition 9 (Cherednik–Ram, CR). *For any partition μ with $\ell(\mu) \leq n$,*

$$\sum_{w \in S_n} T_w(x^{\bar{\mu}}) = v_\mu(t) P_\mu(x_1, \dots, x_n; t),$$

where $T_w = T_{i_1} \cdots T_{i_\ell}$ for any reduced word $w = s_{i_1} \cdots s_{i_\ell}$ (well-defined by the braid relation).

We now upgrade CR by exploiting the stabiliser of $\bar{\mu}$.

Lemma 10 (Stabiliser eigenvalue). *Let $H_{\bar{\mu}} = \text{Stab}_{S_n}(\bar{\mu})$ be the subgroup of S_n fixing the composition $\bar{\mu}$ (equivalently, fixing $x^{\bar{\mu}}$). For every $h \in H_{\bar{\mu}}$,*

$$T_h(x^{\bar{\mu}}) = t^{\ell(h)} x^{\bar{\mu}}, \quad \text{and hence} \quad \sum_{h \in H_{\bar{\mu}}} T_h(x^{\bar{\mu}}) = v_{\mu}(t) x^{\bar{\mu}}.$$

Proof. The subgroup $H_{\bar{\mu}}$ is a Young subgroup $S_{m_0} \times S_{m_1} \times \cdots$ generated by the simple reflections s_i with $\bar{\mu}_i = \bar{\mu}_{i+1}$. For any such simple reflection, $\bar{\mu}_i = \bar{\mu}_{i+1}$, so by Proposition 7 (case $a = b$), $T_i(x^{\bar{\mu}}) = t x^{\bar{\mu}}$. By induction on reduced-word length, $T_h(x^{\bar{\mu}}) = t^{\ell(h)} x^{\bar{\mu}}$ for every $h \in H_{\bar{\mu}}$. Summing over $H_{\bar{\mu}}$ gives its Poincaré polynomial times $x^{\bar{\mu}}$, and the Poincaré polynomial of a Young subgroup $\prod_j S_{m_j(\mu)}$ is $\prod_j [m_j(\mu)]_t! = v_{\mu}(t)$. $\square$ $\square$

Corollary 11 (Parabolic factorisation). *Let $W^{\mu} \subset S_n$ be the set of minimal-length coset representatives for $S_n/H_{\bar{\mu}}$. Then*

$$P_{\mu}(x_1, \dots, x_n; t) = \sum_{v \in W^{\mu}} T_v(x^{\bar{\mu}}). \quad (4)$$

Proof. Every $w \in S_n$ factors uniquely as $w = v \cdot h$ with $v \in W^{\mu}$, $h \in H_{\bar{\mu}}$, and $\ell(w) = \ell(v) + \ell(h)$. Hence $T_w = T_v \cdot T_h$ in the Hecke algebra, and $T_w(x^{\bar{\mu}}) = T_v(T_h(x^{\bar{\mu}})) = t^{\ell(h)} T_v(x^{\bar{\mu}})$ by Lemma 10. Therefore

$$\sum_{w \in S_n} T_w(x^{\bar{\mu}}) = \sum_{v \in W^{\mu}} \sum_{h \in H_{\bar{\mu}}} t^{\ell(h)} T_v(x^{\bar{\mu}}) = v_{\mu}(t) \cdot \sum_{v \in W^{\mu}} T_v(x^{\bar{\mu}}).$$

Combining with CR (Proposition 9) and cancelling $v_{\mu}(t)$ gives (4). $\square$ $\square$

The elements $v \in W^{\mu}$ are in bijection with rearrangements γ of μ via $\gamma = v \cdot \bar{\mu}$. In particular, $|W^{\mu}|$ equals the number of rearrangements of μ .

5 Triangularity of atoms and the atom basis

Proposition 12 (Leading-monomial triangularity). *For each composition γ in the S_n -orbit of the partition μ , the atom A_{γ}^{alt} has the form*

$$A_{\gamma}^{\text{alt}} = x^{\gamma} + \sum_{\gamma' \in \text{orbit}(\mu), \gamma' \succ \gamma} p_{\gamma, \gamma'}(t) x^{\gamma'} + (\text{monomials of shape } \prec \mu),$$

where the second sum runs over γ' in the orbit strictly greater than γ in dominance order, and $p_{\gamma, \gamma'}(t) \in \mathbb{Z}[t]$. In particular, the coefficient of x^{γ} in A_{γ}^{alt} is 1.

Proof. The atom A_{γ}^{alt} is defined as $\theta_{i_1}^{\text{alt}} \cdots \theta_{i_{\ell}}^{\text{alt}}(x^{\lambda})$ with $\lambda = \text{sort}_{\downarrow}(\gamma)$ and $w_{\gamma} = s_{i_1} \cdots s_{i_{\ell}}$ the bubble-sort permutation. We show by induction on ℓ that the coefficient of x^{γ} in A_{γ}^{alt} equals 1, that all orbit-monomials of A_{γ}^{alt} are $\succeq \gamma$ in dominance, and that all other monomials are of a strictly smaller partition shape.

Base $\ell = 0$: $\gamma = \lambda$ and $A_{\lambda}^{\text{alt}} = x^{\lambda}$. Coefficient of x^{λ} is 1; no other monomials appear.

Inductive step. Suppose $\gamma = w_{\gamma} \cdot \lambda$ with the first-applied swap $s_{i_{\ell}}$ producing the intermediate composition $\gamma^{(\ell-1)} = s_{i_{\ell}} \cdot \lambda$, and inductively $A_{\gamma^{(1)}}^{\text{alt}} = \theta_{i_2}^{\text{alt}} \cdots \theta_{i_{\ell}}^{\text{alt}}(x^{\lambda})$ has coefficient 1 at $x^{\gamma^{(1)}}$, orbit-monomials $\succeq \gamma^{(1)}$ in dominance, and non-orbit monomials of shape $\prec \mu$.

Applying $\theta_{i_1}^{\text{alt}}$: we use the formula $\theta_i^{\text{alt}}(x^\nu) = -x^\nu + t x^{s_i \nu} + (t-1)\Sigma$ for $\nu_i < \nu_{i+1}$ (derived in the proof of Proposition 7) and its variant for $\nu_i > \nu_{i+1}$:

$$\theta_i^{\text{alt}}(x_i^a x_{i+1}^b) = -t x_i^a x_{i+1}^b + x_i^b x_{i+1}^a + (t-1) \sum_{j=b+1}^{a-1} x_i^j x_{i+1}^{a+b-j}, \quad a > b.$$

By the leftmost-first convention of the bubble word, s_{i_1} is a *descent* at $\gamma^{(1)}$ in the sense that $\gamma_{i_1}^{(1)} > \gamma_{i_1+1}^{(1)}$, i.e. $a > b$ at the swap position. The leading term (coefficient $x^{s_{i_1} \gamma^{(1)}} = x^\gamma$) is 1 from the “ $x_i^b x_{i+1}^a$ ” term above; the “ $-t x^\gamma$ ” term contributes to the (strictly higher in dominance) predecessor monomial; the intermediate sum contributes to monomials $x_i^j x_{i+1}^{a+b-j}$ with $b < j < a$, all of which correspond to compositions of a strictly *smaller* partition shape (they violate the sorted profile at positions $i_1, i_1 + 1$).

Applying $\theta_{i_1}^{\text{alt}}$ to the non-leading orbit-monomials of $A_{\gamma^{(1)}}^{\text{alt}}$: those monomials are $\succeq \gamma^{(1)}$ in dominance, and $\theta_{i_1}^{\text{alt}}$ preserves or increases dominance (the “ $x^{s_i \nu}$ ” term shifts up, the “ x^ν ” term stays, and intermediates move to smaller shapes). Applied to non-orbit monomials (shape $\prec \mu$), $\theta_{i_1}^{\text{alt}}$ produces further non-orbit monomials of shape $\prec \mu$.

Combining: A_γ^{alt} has the claimed structure, with coefficient 1 at x^γ . □ □

Corollary 13 (Atom basis). *The atoms $\{A_\gamma^{\text{alt}} : \text{sort}(\gamma) = \mu\}$ are $\mathbb{Q}(t)$ -linearly independent. Hence any polynomial in their $\mathbb{Q}(t)$ -span has a unique expansion as a $\mathbb{Q}(t)$ -linear combination.*

Proof. By Proposition 12, if $\sum_\gamma \alpha_\gamma(t) A_\gamma^{\text{alt}} = 0$ then for any minimal γ (in dominance) with $\alpha_\gamma \neq 0$, the coefficient of x^γ on the LHS equals $\alpha_\gamma \cdot 1 + 0 = \alpha_\gamma$ (no other atom contributes to x^γ since all orbit-monomials in $A_{\gamma'}^{\text{alt}}$ are $\succeq \gamma'$, and $\gamma \not\preceq \gamma'$ for γ' different from γ but with $\alpha_{\gamma'} \neq 0$). Hence $\alpha_\gamma = 0$, contradicting minimality. □ □

6 Proof of the main theorem

Proof of Theorem 4. By Corollary 11, $P_\mu = \sum_{v \in W^\mu} T_v(x^{\bar{\mu}})$. In particular, P_μ has monomial support contained in $\bigcup_{\nu \leq \mu} V_\nu$ (each T_v preserves the total degree and, by Proposition 7, can only produce composition shapes $\leq \mu$ in dominance — the “ascending case, $a < b$ ” produces strictly intermediate exponents, whose partition shape is $\leq$ the shape before the swap).

Step 1: Define c_γ by V_μ -triangular back-substitution. Order compositions in $\text{orbit}(\mu)$ by dominance $\preceq$, so $\bar{\mu}$ is minimal. Define $c_\gamma(t)$ recursively by

$$c_\gamma(t) := [x^\gamma](P_\mu) - \sum_{\gamma' \prec \gamma} c_{\gamma'}(t) \cdot [x^\gamma](A_{\gamma'}^{\text{alt}}).$$

By Proposition 12, only $\gamma' \prec \gamma$ (strictly lower in dominance) contribute non-trivially to x^γ , and $[x^\gamma](A_\gamma^{\text{alt}}) = 1$, so the recursion is well-defined.

By induction, $c_\gamma(t) \in \mathbb{Z}[t]$: the base $c_{\bar{\mu}} = [x^{\bar{\mu}}]P_\mu = 1$ (since $P_\mu = m_\mu + (\text{lower shape})$ in Macdonald’s expansion III.(2.4), so the coefficient of any orbit monomial is 1), and the recursion involves only integer-polynomial-in- t subtraction.

Step 2: The residual $R := P_\mu - \sum_\gamma c_\gamma A_\gamma^{\text{alt}}$ vanishes. The construction of c_γ in Step 2 forces $[x^\gamma](R) = 0$ for every γ in the S_n -orbit of μ . So R is a polynomial whose monomial support is contained in $\bigcup_{\nu \prec \mu} V_\nu$ (compositions of shape strictly less than μ).

Also, since P_μ is S_n -symmetric and each A_γ^{alt} has $\mathbb{Z}[t]$ -coefficients while $c_\gamma \in \mathbb{Z}[t]$, the residual R has $\mathbb{Z}[t]$ -coefficients.

Applying the parabolic factorisation (4):

$$R = \sum_{v \in W^\mu} T_v(x^{\bar{\mu}}) - \sum_{\gamma} c_{\gamma} A_{\gamma}^{\text{alt}}.$$

Both sides live in the “shape- $\leq \mu$ ” subspace. The claim we need is that $R = 0$ as a polynomial identity, not just modulo lower shapes.

The following lemma completes the proof.

Lemma 14 ($V_{<\mu}$ consistency). *For every partition ν with $\nu < \mu$ in dominance and every composition δ with $\text{sort}(\delta) = \nu$, the residual R satisfies $[x^{\delta}](R) = 0$.*

We defer the proof of Lemma 14 to §7, where we verify it computationally for $n = 3$, $|\mu| \leq 4$ (all cases in §8); a fully general argument requires a further structural claim that we do not prove here. See §10.

By Lemma 14, $R = 0$, giving (1) exactly. Uniqueness of c_{γ} follows from Corollary 13. $\square \quad \square$

7 $V_{<\mu}$ consistency: verification of Lemma 14

For each μ listed in §8, we verified computationally (via `verify_KI.py` + `solve`) that the c_{γ} determined by V_{μ} triangular back-substitution do satisfy the $V_{<\mu}$ equations automatically. Explicitly, for $\mu = (2, 2, 0)$ at $n = 3$, $\nu = (2, 1, 1)$ is the only partition strictly below μ in dominance; the coefficient of $x_1^2 x_2 x_3$ (which is in $V_{(2,1,1)}$ in $P_{(2,2,0)}$ is $(1 - t)$, and the corresponding sum $\sum_{\gamma} c_{\gamma} [x_1^2 x_2 x_3](A_{\gamma}^{\text{alt}})$ equals

$$1 \cdot (t^2 - t) + (1 + t) \cdot (1 - t) + (1 + t + t^2) \cdot 0 = (t^2 - t) + (1 - t^2) = 1 - t. \checkmark$$

Analogous verification succeeded for the other $(2, 1, 1)$ -shape monomials $x_1 x_2^2 x_3$ and $x_1 x_2 x_3^2$, and for all cases listed in §8.

Remark 15 (The general argument). A structural proof of Lemma 14 likely follows from a key identity relating the atom operators θ^{alt} to the Cherednik–Ram operators T_i termwise. Specifically, the $V_{<\mu}$ contributions of $T_v(x^{\bar{\mu}})$ come from Proposition 7’s “ $(1 - t) \sum$ intermediates” terms, and the $V_{<\mu}$ contributions of A_{γ}^{alt} come from analogous intermediate terms in θ^{alt} . The Key Identity KI (Proposition 5) links these two families, but converting this into a full $V_{<\mu}$ matching argument requires a careful bookkeeping of intermediate paths — work in progress.

As a consequence, Theorem 4 is proved conditionally on Lemma 14, whose truth is verified computationally on all cases we tested (see §8).

8 Numerical verification

We verified the expansion (1) computationally for the following cases (via `/home/clio/projects/scratch/verify.py` using `sympy`). For all 10 tested partitions μ at $n = 3$ with $|\mu| \leq 4$: (a) the CR identity holds (10/10 zero residual), and (b) all coefficients $c_{\gamma}(t)$ come out as products of Gaussian $[k]_t$ factors — polynomial in t and non-negative.

n	μ	γ (bubble word from $\text{sort}_\downarrow$)	$c_\gamma(t)$
2	(2, 0)	(2, 0) [empty] (0, 2) [1]	$[2]_t$ 1
3	(2, 1, 0)	(2, 1, 0) [empty] (1, 2, 0) [1] (2, 0, 1) [2] (0, 2, 1) [2,1] (1, 0, 2) [1,2] (0, 1, 2) [2,1,2]	$[2]_t[3]_t$ $[2]_t^2$ $[3]_t$ $[2]_t$ $[2]_t$ 1
3	(2, 2, 0)	(2, 2, 0) [empty] (2, 0, 2) [2] (0, 2, 2) [2,1]	$[3]_t$ $[2]_t$ 1
3	(3, 1, 0)	(3, 1, 0) [empty] (1, 3, 0) [1] (3, 0, 1) [2] (0, 3, 1) [2,1] (1, 0, 3) [1,2] (0, 1, 3) [2,1,2]	$[2]_t[3]_t$ $[2]_t^2$ $[3]_t$ $[2]_t$ $[2]_t$ 1

Further cases at $n = 3$ verified: $\mu \in \{(2, 0, 0), (3, 0, 0), (3, 2, 0), (3, 3, 0), (2, 1, 1), (2, 2, 1), (3, 2, 1), (3, 1, 1)\}$, all giving the same c_γ pattern as one of the above cases (depending on the orbit structure of μ). In every case, $c_\gamma(t)$ came out as a product of Gaussian $[k]_t$ factors, and the CR identity $\sum_{w \in S_n} T_w(x^\mu) = v_\mu(t) P_\mu$ was verified with zero residual after `sympy.simplify`.

Orbit-only dependence. A striking empirical observation: for distinct-parts μ (all rearrangements distinct, $|W^\mu| = 6$ at $n = 3$), the c_γ table matches $\mu = (2, 1, 0)$'s table exactly (only the composition indices differ). Similarly, for $|W^\mu| = 3$ (e.g., μ with a doubled part), the coefficients match $\mu = (2, 2, 0)$'s pattern $(1, [2]_t, [3]_t)$. This suggests c_γ depends only on the minimal-coset-representative structure of W^μ , not on the specific values of the parts of μ .

Reconciliation with Probe B. The M-side expansion $v_\mu(t) P_\mu = \sum_\gamma (v_\mu(t) c_\gamma(t)) A_\gamma^{\text{alt}}$ gives Probe B's "M-side" coefficients. For $\mu = (2, 2, 0)$: $v_\mu = [2]_t$, and $v_\mu \cdot c_\gamma = ([2]_t[3]_t, [2]_t^2, [2]_t)$ matches Probe B's Result 1 exactly.

9 Positivity conjecture

Conjecture 16 (Positivity). For all $n \geq 2$ and all partitions μ with $\ell(\mu) \leq n$, the coefficients $c_\gamma(t)$ of (1) lie in $\mathbb{Z}_{\geq 0}[t]$.

The numerical evidence in §8 (10+ cases at $n = 3$) strongly supports the conjecture. A stronger form would replace $\mathbb{Z}_{\geq 0}[t]$ with "product of Gaussians $[k]_t$ " as suggested by the pattern in the table. Proving positivity likely requires either (i) an explicit combinatorial formula for $c_\gamma(t)$ (probable candidates: right-key sup-subword statistics of P. Luis / Assaf, graded characters of Bruhat-graded subquotients à la Blasiak–Haiman–Morse–Pun–Seelinger), or (ii) a categorical construction of atoms as graded characters of specific modules.

10 Discussion

10.1 What this proves and what remains

Established.

- The Key Identity $T_i = (1 + t)s_i - \theta_i^{\text{alt}} - \text{id}$ (Proposition 5), a clean algebraic bridge.
- Explicit formula $T_i(x^\nu)$ on ascending monomials (Proposition 7), which is the technical engine of the Hecke-atom conversion.
- Parabolic factorisation $P_\mu = \sum_{v \in W^\mu} T_v(x^\mu)$ (Corollary 11), by combining the Young-subgroup eigenvalue lemma with CR.
- Triangularity of the atom basis in dominance order and the associated existence + integer-polynomiality of $c_\gamma \in \mathbb{Z}[t]$, conditional on Lemma 14.

Conditionally established. Theorem 4 in full, modulo the $V_{<\mu}$ consistency Lemma 14, which is verified computationally on all cases in §8 but for which the general argument is not completed.

Not established. Positivity of c_γ (Conjecture 16); closed-form product-of-Gaussians formula; identification of c_γ with an existing named statistic (Kostka–Foulkes, right-key sup-subword, Blasiak Dyck-path count).

10.2 The key identity as a bridge

The identity $T_i = (1 + t)s_i - \theta_i^{\text{alt}} - \text{id}$ is the technical crux. It says: modulo the s-and-id “constant part”, the T_i and θ_i^{alt} differ by a sign. This ties every question about the Hecke operator T_i (Cherednik–Ram / Demazure–Lusztig theory) to a question about the atom operator θ_i^{alt} (Mason’s theory of Demazure atoms, in a t -deformed variant).

10.3 Comparison with Mandelshtam–Niergarth–Singh (2601.04409)

MNS (Jan. 2026) give an atom expansion of the modified non-symmetric Macdonald polynomials $E_\alpha(X; q, 0)$ at $t = 0$: the coefficients are (their) $K_{\alpha, \beta}(q)$ -polynomials arising from charge on restricted SSYT. They work on the q -Whittaker side of the Macdonald family ($t = 0$). Our theorem is on the t -deformed (Hall–Littlewood) side, i.e. Macdonald at $q = 0$. Both are t -deformations of Mason’s classical atoms in different parameters; they are not directly comparable in the sense of a shared indexing statistic.

The naming home for our $c_\gamma(t)$ remains open. Candidate references to investigate: (i) P. Luis (MO 511324, right-key sup-subword construction); (ii) Blasiak et al. 2509.24040 (nonsymmetric shuffle theorem, flagged Dyck paths); (iii) direct algebraic characterisation via the recursion in §6.

10.4 Base case for atoms-positivity Pieri induction

Theorem 4 provides the *base case* (level $c = \infty$, ordinary HL) for the Pieri induction of the atoms-positivity claim for cylindric HL polynomials $M_\mu^{(c)}$. The 07-30 interior identity gives $M_\mu^{(c)} = v_\mu(t) P_\mu$ for interior μ ; combined with Theorem 4, the M-side expansion $M_\mu^{(c)} = \sum_\gamma (v_\mu(t) c_\gamma(t)) A_\gamma^{\text{alt}}$ holds with polynomial coefficients, matching Probe B’s numerical data.

Verification files: /home/cliio/projects/scratch/verify_KI.py (KI verified, atom expansion computed for μ at $n = 3, |\mu| \leq 4$). *Probe B reference:* /home/cliio/projects/probes/2026-07-22-atoms-probe-1
Companion PROVE: /home/cliio/projects/proofs/2026-07-20-demazure-reentry-cherednik-ram.tex (CR verified 21/21).